

Downlink-Assisted Uplink Beam Steering in an Asymmetric Underwater Optical Attocell for Turbulence-Resilient Links

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Abstract *An asymmetric underwater optical attocell is proposed to mitigate the coverage-efficiency trade-off in underwater wireless optical communication. Downlink diffusive beams enable extended cell formation, while uplink beam steering exploits positional information. Experiments demonstrate turbulence-resilient coverage and stable communication throughout the attocell for mobile nodes. ©2026 The Author(s)*

Introduction

In Beyond 5G and 6G communication systems, non-terrestrial and underwater environments are expected to support applications such as underwater sensing systems and autonomous vehicles, which require high data rates and low latency [1]. Conventional underwater acoustic communication is limited by narrow bandwidth and high latency, while radio frequency signals experience severe attenuation in seawater. As a result, underwater wireless optical communication (UWOC) in the blue-green spectral region has emerged as a promising alternative [2-4].

However, UWOC is highly sensitive to misalignment and turbulence-induced fluctuations in intensity due to its strong directionality, making reliable communication with mobile nodes challenging [5-7]. Similar turbulence-induced fading and link instability have also been reported in free-space optical communication systems [8]. Although wide field-of-view receivers and diffusive beam transmission improve link acquisition, they reduce optical power density and introduce a fundamental trade-off between coverage and communication efficiency.

This trade-off becomes particularly critical in cell-based optical communication. Optical attocells have been widely studied for indoor visible light communication, where dense deployment enables seamless connectivity [9-12]. In underwater environments, however, absorption, scattering, and turbulence make it difficult to achieve stable, uniform coverage regions. In addition, conventional bidirectional optical communication systems often rely on asymmetric wavelengths but employ similar beam characteristics in both directions, which does not fundamentally resolve the trade-off.

To address this challenge, this study proposes an asymmetric underwater optical attocell (UOA) based on directional separation. The downlink employs a wide diffusive beam to achieve robust coverage, while the uplink uses a narrow beam with beam steering to enhance received power and support efficient transmission.

By decoupling design requirements for each transmission direction, the proposed approach mitigates the inherent trade-off in conventional systems. The effectiveness of the proposed concept is validated through experimental demonstrations of the downlink and uplink functions, enabling quantitative evaluation of the dominant impairments in each direction. These experiments evaluate complementary aspects of downlink coverage and uplink alignment performance.

System Design of Underwater Optical Attocell

In UWOC, there exists an inherent trade-off between beam divergence and received optical power. A wide beam improves coverage and link acquisition for mobile nodes, but reduces optical power density due to spatial spreading and attenuation in water. In contrast, a narrow beam provides high optical power density and enables

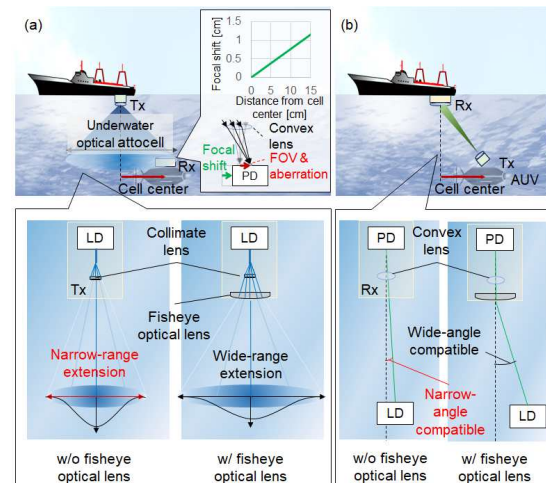


Fig. 1: Concept of the proposed asymmetric UOA: (a) downlink diffusion and (b) uplink beam steering.

Tab. 1: Requirements and design guidelines.

	Requirement	Design
Downlink	Coverage, low rate	Wide diffusive beam, low power density
Uplink	High rate, stability	Narrow beam, steering, high power density
Proposed	Coverage and efficiency	Asymmetric design, DL-assisted steering

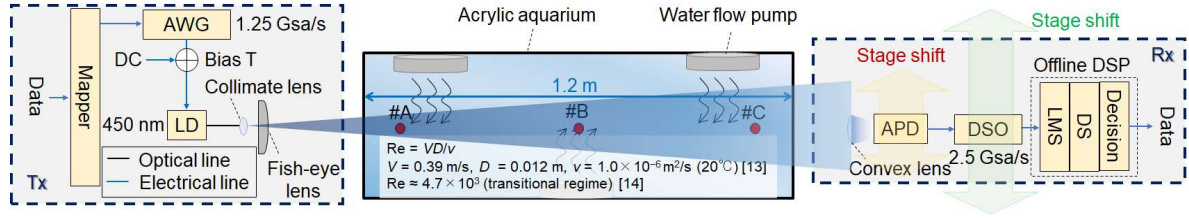


Fig. 2: Experimental setup for downlink beam diffusion.

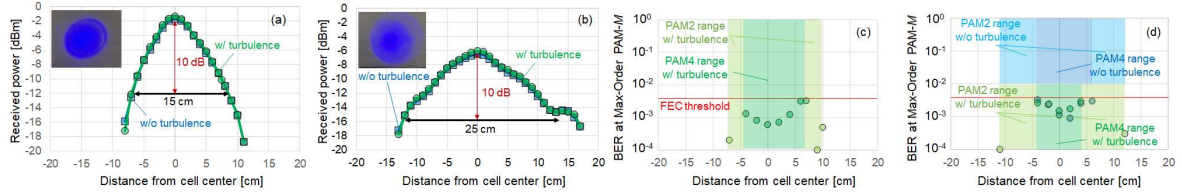


Fig. 3: Downlink performance: received power distributions for beam diameters of (a) 15 cm and (b) 25 cm, and corresponding BER distributions under (c) turbulence and (d) without turbulence.

efficient transmission, but is highly sensitive to misalignment and turbulence, making stable link maintenance difficult. As a result, achieving both wide coverage and high communication efficiency with a single beam configuration is fundamentally challenging.

Conventional bidirectional UWOC systems typically employ asymmetric wavelengths while maintaining symmetric beam configurations. However, such designs cannot resolve the conflicting requirements between coverage and efficiency, particularly in mobile scenarios where alignment and channel conditions vary dynamically. To address this issue, we propose an asymmetric UOA based on an asymmetric beam design, in which the uplink and downlink are assigned distinct and complementary roles. Fig. 1(a) and (b) illustrate the proposed system. A fisheye lens is deployed at the base station to provide a wide field of view, enabling robust reception over a wide angular range and facilitating link formation within the attocell.

In the downlink, robust link acquisition and wide-area coverage are the primary requirements. Therefore, a wide diffusive beam with low optical power density is employed to support link formation. The beam diameter can be controlled to adjust the cell size, allowing a trade-off between coverage and the distribution of received power. Furthermore, due to the fisheye lens's wide field of view and the convex lens's optical aberrations, the position of the incident light on the photodetector varies with the angle of incidence, providing implicit information about the receiver's position within the cell.

This positional information is used to determine the uplink beam-steering angle, thereby significantly reducing the search space required for alignment. In the uplink, a narrow beam with high optical power density is used to achieve high-efficiency transmission. The transmission direction

is optimized based on downlink-derived position information, enabling effective alignment even in the presence of mobility and turbulence.

By separating design requirements by transmission direction and linking the two directions via positional information, the proposed optical attocell overcomes the fundamental limitations of conventional symmetric systems, enabling robust coverage and efficient communication.

Tab. 1 summarizes the correspondence between system requirements and design guidelines for each transmission direction.

Downlink Beam Diffusion: Results

Fig. 2 shows the experimental setup for the downlink. At the transmitter, a signal generated by an arbitrary waveform generator (AWG) and a mapper at a symbol rate of 312.5 Msymbol/s was applied to a 450 nm laser diode (LD) via a bias-tee, employing pulse amplitude modulation with M levels (PAM- M , $M = 2$ and 4). The emitted light was collimated by a collimation lens and a fisheye lens to form a wide, diffuse beam. The collimation lens adjusted the beam diameter, and multiple beam sizes were evaluated to investigate the relationship between coverage and received power distribution.

At the receiver, the collected light was focused by a convex lens and detected by an avalanche photodiode (APD). The signal was captured with a digital storage oscilloscope (DSO) and processed offline with least-mean-square (LMS) equalization, downsampling (DS), and decision processing. The receiver was mounted on a stage to enable horizontal movement. A water tank served as the propagation channel, and a pump generated turbulence. The points labeled #A to #C indicate representative positions where turbulent conditions were confirmed, with Reynolds numbers exceeding 4000.

Fig. 3 shows the distributions of received

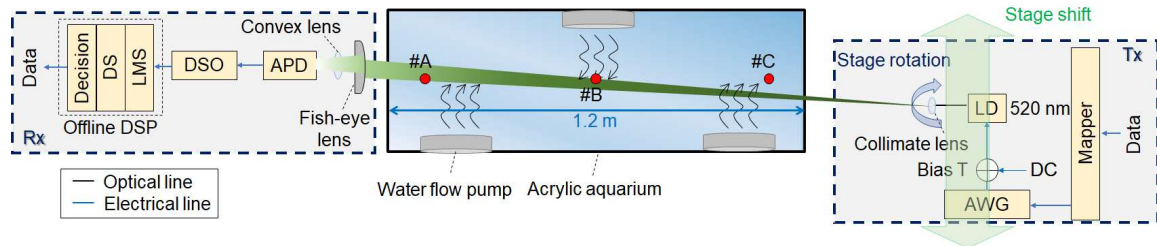


Fig. 4: Experimental setup for uplink beam steering.

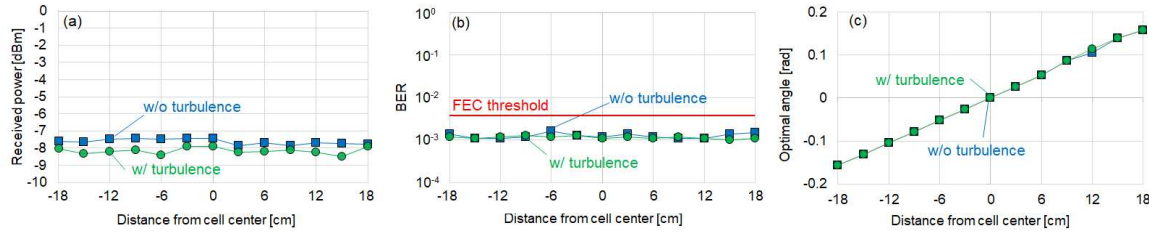


Fig. 5: Uplink performance: (a) received power, (b) BER, and (c) optimal steering angle.

power and bit error ratio (BER). Figs. 3(a) and 3(b) present the received power distributions for beam diameters of 15 cm and 25 cm, respectively. For the 15 cm beam, the received power is concentrated near the center with a width of approximately 15 cm, whereas the 25 cm beam provides a wider distribution covering approximately 25 cm. Notably, the peak power and overall distribution remain nearly unchanged under turbulence, indicating that diffusive transmission mitigates power degradation. This is attributed to the spatial averaging of intensity fluctuations arising from multiple propagation paths.

Figs. 3(c) and 3(d) show the BER distributions under turbulent and non-turbulent conditions. Based on the forward error correction (FEC) threshold, the communication region expands as the beam diameter increases. While higher-order modulation is limited to the central region, lower-order modulation enables reliable communication over a wider area. Moreover, even in the presence of turbulence, the BER remains below the threshold over a large region, demonstrating the robustness of the diffusive beam.

Uplink Beam Steering: Results

Fig. 4 shows the experimental setup for the uplink. A narrow beam from a 520 nm LD was used at the transmitter, and the transmission angle was controlled using a rotational stage to maximize received power, with PAM4 modulation. The receiver configuration, signal processing, and turbulence generation are the same as in Section 3, except that a fisheye lens is used to capture light over a wide field of view. The points labeled #A to #C indicate representative positions where turbulent conditions were confirmed with Reynolds numbers exceeding 4000.

Fig. 5(a) shows the spatial distribution of

received power. The received power varies with the transmission angle and exhibits a clear peak at the optimal geometric alignment between the transmitter and receiver, reflecting the narrow beam's high directivity. The overall distribution is preserved under turbulence, indicating that the transmission direction primarily determines the received power.

Fig. 5(b) presents the BER distribution. The minimum BER is achieved at the angle corresponding to the maximum received power, while it degrades rapidly as the transmission angle deviates from the optimum. Nevertheless, the BER remains below the FEC threshold across the evaluated angles. This result indicates that alignment accuracy is the dominant factor in uplink performance. Even under turbulent conditions, no significant degradation is observed in the direction of maximum gain, demonstrating that stable communication can be maintained through beam steering.

Fig. 5(c) shows the optimal transmission angle with respect to the cell center. A monotonic relationship between the receiver position and the optimal angle is observed, indicating a one-to-one mapping. This enables determining the appropriate transmission direction at any position within the cell based on the received power. Furthermore, this relationship is maintained under turbulence, confirming that the proposed method is robust to environmental fluctuations.

Conclusions

An asymmetric UOA was experimentally demonstrated to resolve the coverage–efficiency trade-off in underwater optical communication. Downlink diffusive beams enable robust cell formation, while downlink-assisted beam steering achieves efficient uplink alignment and maximizes received power.

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